Task 25

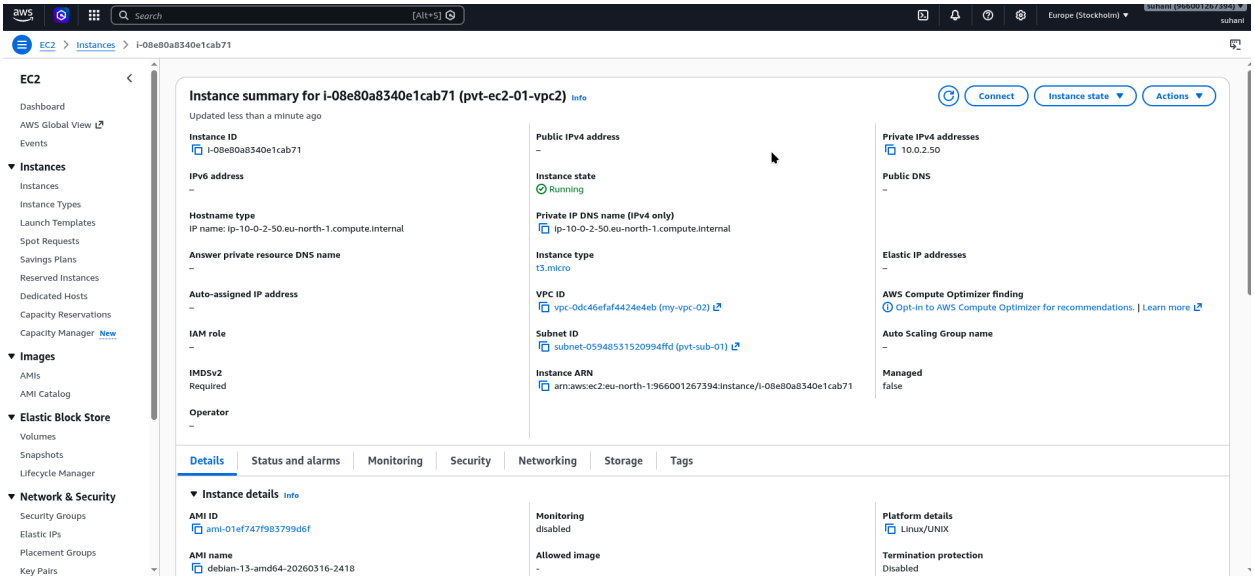
Implement Private EC2 with S3 Access via VPC Endpoint
Create Gateway VPC Endpoint for S3
Remove NAT Gateway
Ensure private EC2 can:

- Access S3
- Cannot access internet
  Validate using CLI

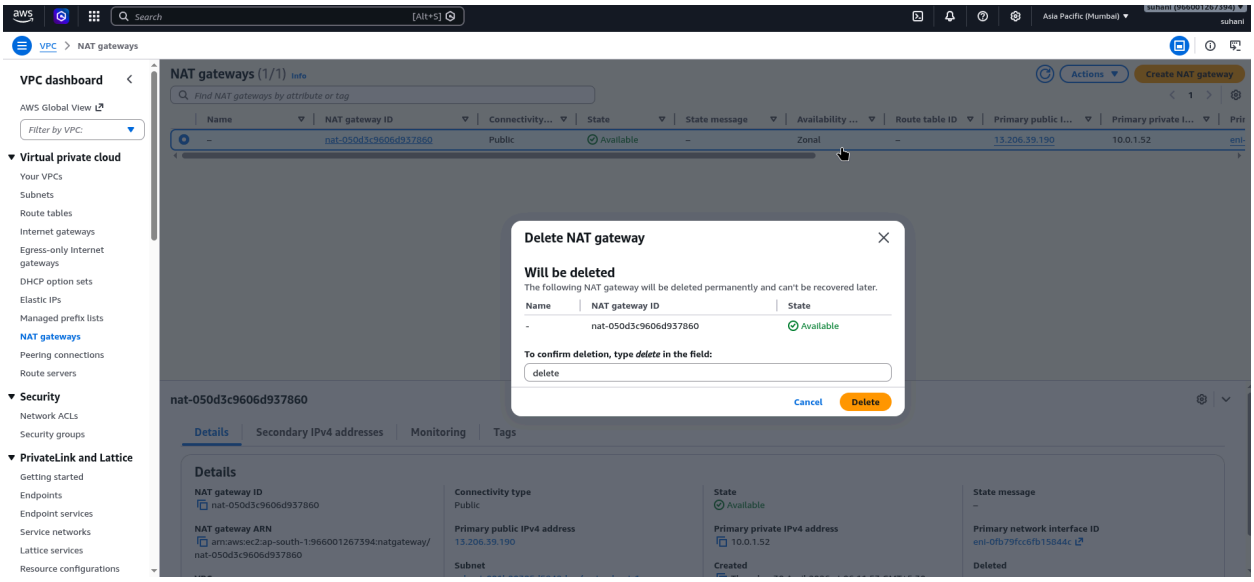
Step 1: Ensure Private EC2

Your EC2 must:

- Be in **private subnet**



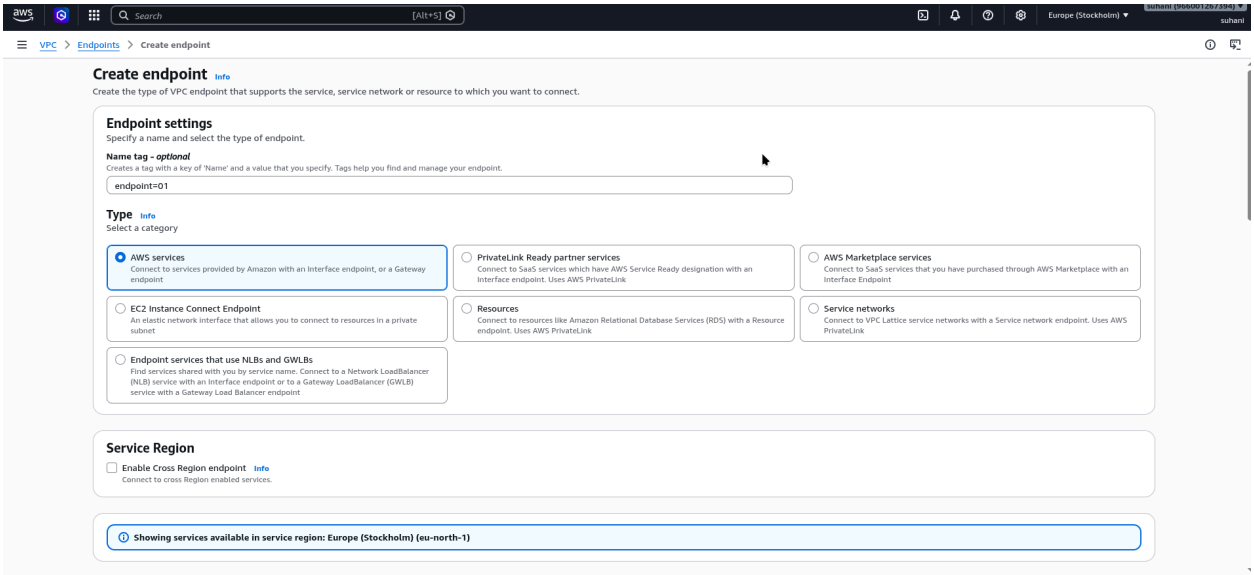
private ec2 in a private subnet



removing nat gateway

Create VPC Endpoint (S3)

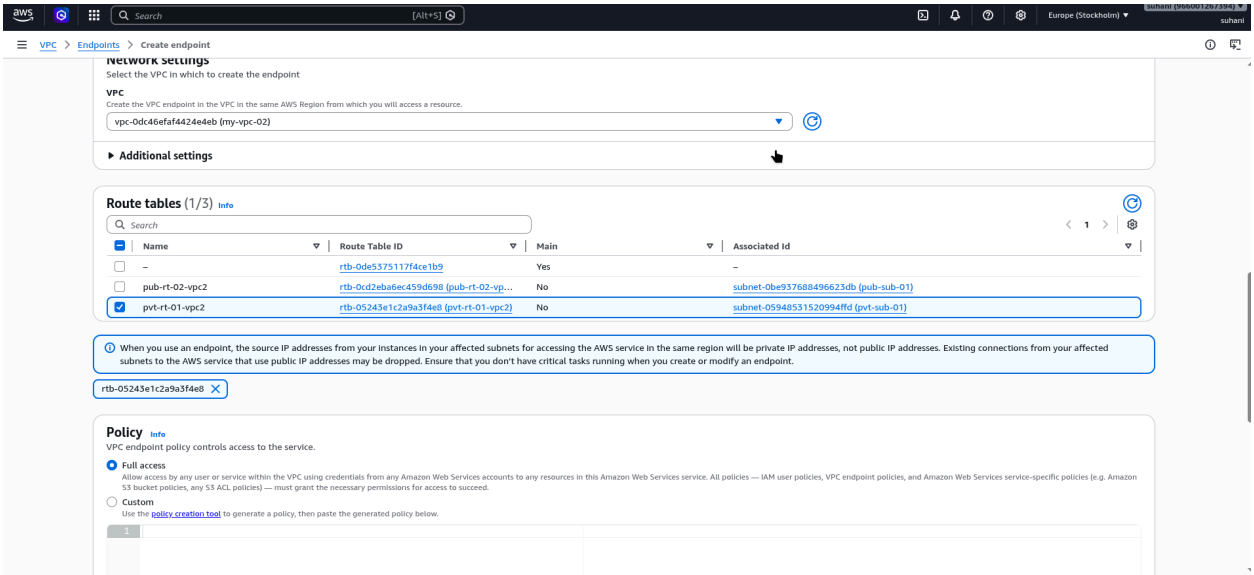
1. Go to → **Endpoints**
2. Click **Create endpoint**



Configure:

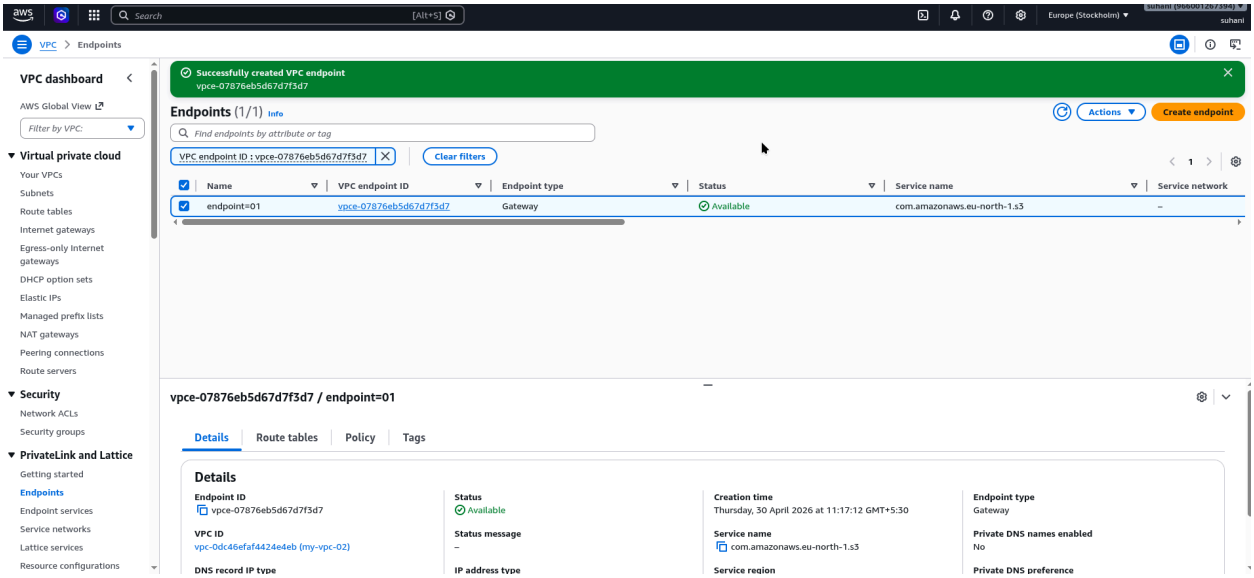
- Service category → AWS services

- Service →
- Type → **Gateway**



Attach:

- Select your **VPC**
- Select **Route Table of private subnet**



AWS adds route:

```
pl-xxxx (S3 prefix list) → vpce-xxxx
```

👉 This allows:

✔️ Private EC2 → S3

❌ No internet

Step 5: Validate using CLI

SSH into your private EC2:

```
suhani@debian:~$ ssh -i "awskey.pem" admin@13.60.69.11
Linux ip-10-0-1-87 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Apr 29 20:40:25 2026 from 152.58.119.218
admin@ip-10-0-1-87:~$ ssh -i "awskey.pem" admin@10.0.2.50
Linux ip-10-0-2-50 6.12.74+deb13+1-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-2 (2026-03-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Apr 25 16:46:14 2026 from 10.0.1.87
admin@ip-10-0-2-50:~$ aws s3 ls
```

```
admin@ip-10-0-2-50:~$ aws s3 ls
2026-04-30 01:50:23 bucket-unique2121-21
2026-04-30 04:36:34 destination-bucket-cheesecake
2026-04-30 02:40:29 my-logs-bucket-0123
2026-04-30 03:11:56 my-static-website-hehw34
2026-04-30 04:43:26 source-bucket-cheese
2026-04-30 00:05:48 test-bucket-93898
2026-04-30 02:35:06 unique-mybucket-12
2026-04-30 02:23:49 versioning-practivce-01
admin@ip-10-0-2-50:~$ ping google.com
```

aws configure and s3 working

```
admin@ip-10-0-2-50:~$ ping google.com
PING google.com (216.58.201.174) 56(84) bytes of data.
^C
--- google.com ping statistics ---
14 packets transmitted, 0 received, 100% packet loss, time 13300ms

admin@ip-10-0-2-50:~$ aws s3 ls
2026-04-30 01:50:23 bucket-unique2121-21
2026-04-30 04:36:34 destination-bucket-cheesecake
2026-04-30 02:40:29 my-logs-bucket-0123
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2026-04-30 02:35:06 unique-mybucket-12
2026-04-30 02:23:49 versioning-practivce-01
admin@ip-10-0-2-50:~$
```

google ping is not working